

WASP-2b

R-1042

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_151103 · created 2026-07-30T08:10:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457329.6054 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1603 +/- 0.0077
Transit depth (Rp/Rs)^2	2.57 +/- 0.25 [%]
Orbital Inclination (inc)	85.51 +/- 0.6
Airmass coefficient 1 (a1)	1.097 +/- 0.01
Airmass coefficient 2 (a2)	-0.063 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	#2 - [64, 226]
Optimal Aperture	3.2
Optimal Annulus	3.46
Transit Duration (day)	0.0702 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1603 ± 0.0077 vs 0.1326 (deviation 3.55σ)
Duration fit vs published	0.0702 d vs 0.0737 d (deviation 1.03σ)
Mid-time O–C	-9.0 min
Depth SNR	20.6
Residual scatter	0.95 %

Light curve

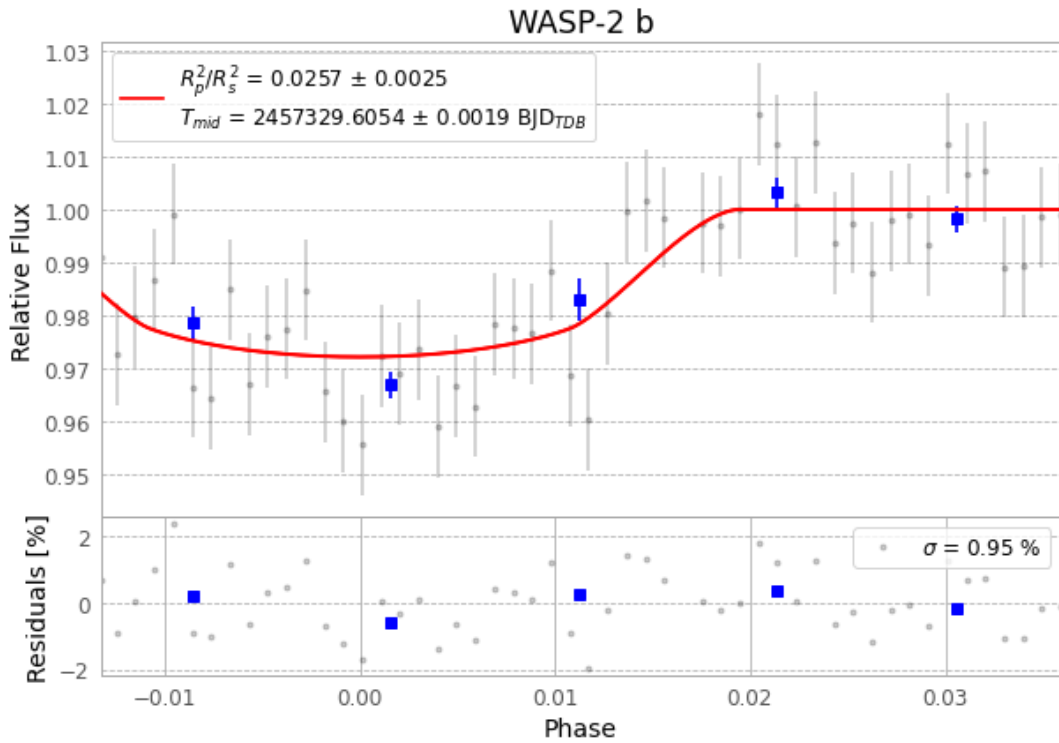


Figure 1 — FinalLightCurve_WASP-2 b_2015-11-02.png (SHA-256 f20c582915b92fa5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 354], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b39c20e2cee903b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

52 FITS frames (34 MB), WASP-2151103014843.FITS → WASP-2151103042112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-151103.log (0cdf6a0792b0...), FinalLightCurve_WASP-2 b_2015-11-02.png (f20c582915b9...), FinalLightCurve_WASP-2 b_2015-11-02.csv (77c10a6658f3...)

Compute resources

Wall time 230.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 08:10Z.